

StudyMate Requirements Documents

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1. Introduction

StudyMate is a standalone desktop application designed to optimize self-study by providing an adaptive scheduling framework. The app bridges the gap between static calendars and actual progress by analyzing user performance and recalibrating future study templates based on real-world completion rates.

2. Target User

- a. **The Student:** Users with fixed deadlines (exams, finals) who need to fit a specific volume of material into a shrinking window of time.
- b. **The Hobbyist:** Users pursuing long-term, self-paced learning with no fixed end date, requiring help with consistency and momentum.

3. Functional Requirement

3.1 Course & Plan Creation

- i. **Standard Input:** The user shall create a course by providing an ISBN (fetches data via API) or by manually defining units (chapters/pages).
- ii. **OCR Input (Stretch Goal):** The user may provide an image of a Table of Contents; the app shall extract text to generate the course structure.
- iii. **Granular Scheduling:** The app shall generate a daily study plan based on user-defined parameters (target end date, daily capacity, or "rest days").

3.2 Progress Tracking & Calibration

- iv. **Active Tracking:** The app must provide a daily "Agenda" where users check off completed units.
- v. **Threshold-Based Updates:** The user shall set a "delay threshold" (e.g., 3 days behind). If exceeded, the app must prompt the user to recalibrate the remaining plan.
- vi. **Post-Plan Feedback:** After a plan concludes, the user shall provide qualitative feedback (difficulty, satisfaction) to inform future templates.

3.3 Reporting & Notifications

- vii. **Progress Visualization:** The app shall provide weekly visual reports comparing "Planned vs. Actual" progress.
- viii. **Desktop Reminders:** The app shall trigger system-level notifications for the daily agenda at a user-specified time.

4. Non-functional Requirement

- a. **Local-First Architecture:** All user data, plans, and history must be stored in a local file. No remote server is required.
- b. **Offline Functionality:** The app must be fully functional without internet access (excluding the optional ISBN lookup).
- c. **Data Portability:** The user shall be able to export their database for backup or manual migration.